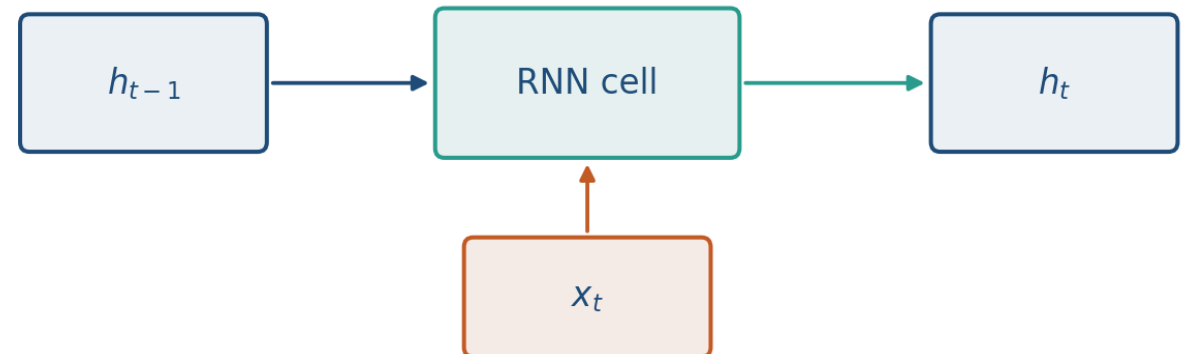


# A cell with memory

- Each step:  $x_t$  and  $h_{t-1}$  become  $h_t$ .
- The same weights at every token.
- Reset the state between unrelated documents.

Same weights at every step: input plus previous state



# Unrolled in time

- The loop is a chain. That chain is the long-range path.
- A CNN is parallel with a fixed window. An RNN is sequential with the whole past.

Unrolled in time. The chain is why long-range gradients struggle.

